OFFICIAL CURRICULUM & PYQ REVIEW

Bihar STET Paper II: Computer Science

Detailed Structural Analysis & Syllabus Segregation (Higher Secondary / Class 11–12)

EXAM PATTERN	NEGATIVE MARKING	ANALYZED PAPERS	TARGET STANDARD
150 Questions | 150 Marks | 150 Mins	**None (1 Mark Per Question)**	**2020, 2023, 2024, 2025 (Multiple Shifts)**	**Senior Secondary Teacher (PGT)**

100 Qs
Part I: Computer Science
Core B.Tech/MCA fundamentals, Emerging Technologies, and Programming.

30 Qs
Part II: Art of Teaching
Pedagogy, Bloom's Taxonomy, Lesson Planning, and Evaluation schemes.

20 Qs
Part III: General Skills
Mental Reasoning, Arithmetic Aptitude, Indian History, and Environmental GK.

Part I: Computer Science Weightage Distribution (100 Marks)

Empirical PYQ Distribution

Subject Domain	Approx. Weight	Primary Focus Areas Identified in Question Papers
Data Structures & Algorithms	**18 – 20%**	Recurrence relations, Sorting ($O(n \log n)$ vs $O(n^2)$), Threaded Trees, Graphs (Prim/Cut property, Floyd-Warshall), Hashing.
Computer Networks & Security	**15 – 18%**	OSI/TCP-IP models, ARP, IP Addressing (IPv4/IPv6), Media types (UTP Cat-5/6), Firewalls, Public-Key Cryptography, Network Sockets.
Programming (OOP & Python/C++)	**12 – 15%**	OOP pillars (Overloading/Overriding, Abstract classes), Python indexing/slicing, dictionaries, memory identity (`is`), C++ I/O.
Database Management Systems	**10 – 12%**	Three-tier abstraction, Relational keys, Locks & concurrency, Normalization, SQL DDL/DML, Aggregation (`HAVING` vs `WHERE`).
Digital Logic & COA	**10 – 12%**	K-Maps (SOP/POS), Logic families (TTL/CMOS), Carry-lookahead adders, Microprocessor registers (Accumulator), Bus width.
Emerging Tech (IoT, AI, E-Com)	**10 – 12%**	IoT sensors, ZigBee, SCADA, BLE, RFID (MISO pin), AI state search (A*, Uniform Cost), E-Commerce models (B2B, Reverse Auction).
Theory of Computation (TOC)	**6 – 8%**	Chomsky Hierarchy, DFA state minimization, CFG closure properties, Pushdown Automata, Turing Machine halting states.
Web Tech & Software Engineering	**6 – 8%**	HTML/CSS media queries, DOM manipulation, SDLC models (Spiral, Waterfall), Estimation metrics (LOC, FP, COCOMO).

Strategic Insight: Bihar STET prioritizes clean theoretical definitions and structural properties over complex coding. More than 45% of technical questions originate directly from foundational core CS topics (DSA, Networking, and DBMS).

1. Digital Logic & Computer Organization

~10–12 Questions

Boolean Algebra, K-Maps & Circuits

High Frequency

- **K-Map Simplification:** 3-variable and 4-variable function minimization; deriving both SOP and POS equations; grouping adjacent cells and the role of "Don't Care" conditions.
- **Logic Families:** Operating voltages of TTL (5V logic); CMOS fabrication characteristics utilizing depletion and enhancement mode MOSFETs.
- **Adders & Converters:** Full adder 3-input requirements; propagation delay advantages of Carry Look-Ahead Adders over Ripple Carry Adders; Priority Encoders.
- **Number Representations:** 1's complement ($[-2^{n-1}, 2^{n-1}-1]$) and 2's complement ranges; handling carry-out from MSB; Excess-3 as a self-complementing code; Gray to Binary conversion using XOR gates.

Computer Architecture & Microprocessors

Conceptual

- **Registers & Buses:** Function of Accumulator register in ALU operations; Data bus width measured in bits; Program Counter and Instruction registers.
- **Microcontrollers:** Harvard architecture (separate memory space for instruction and data) vs. Von Neumann; Memory-Mapped I/O mechanics; automotive embedded usage.

2. Data Structures & Algorithms (DSA)

~18–20 Questions

Algorithmic Complexity & Paradigms

Core Analytical

- **Asymptotic Analysis:** Solving divide-and-conquer recurrences ($T(n)=T(n/2)+1$ yielding $O(\log n)$); conditional asymptotic notation ($o(f(n))\text{ if }g(n)$); definition of NP-completeness and polynomial-time verification.
- **Paradigms:** Divide & Conquer (recursive subproblem breakdown; finding min-max in $O(n)$); Dynamic Programming (optimal substructure, memoization); Backtracking (8-Queens, Hamiltonian Cycle).

Linear & Non-Linear Data Structures

Heavily Repeated

- **Linear Structures:** Stack applications (DFS traversal, recursion elimination using iterative loops, postfix evaluation); 2-D array row-major addressing formula `Index = B + (i*n + j)`; constant-time $O(1)$ operations in linked lists.
- **Trees:** Threaded binary trees (predecessor/successor pointer replacement); AVL tree balance factor ($-1, 0, +1$); B-tree order m child capacities; BST single-child node deletion mechanics.
- **Graphs:** Spanning trees (minimum $n-1$ edges for n vertices); Prim's cut property; Floyd-Warshall all-pairs shortest paths ($O(n^3)$).
- **Sorting & Hashing:** Time complexities (Merge Sort $O(n \log n)$, Insertion Sort $O(n^2)$); Counting Sort constraints; Hash collisions (Separate Chaining via linked lists vs. Linear Probing).

3. Database Management Systems (DBMS)

~10–12 Questions

Architecture, Relational Model & SQL

Standard Theory

- **Architecture:** Three levels of data abstraction (Physical, Conceptual/Logical, View); Inodes; Master/Perpetual files; Concurrency control using Locks.
- **Relational Keys:** Candidate keys, Primary keys, Foreign keys; Clustered indices (data record order matching index order).
- **SQL & Normalization:** DDL (`DROP`, `CREATE`) vs. DML (`UPDATE`, `INSERT`); `HAVING` vs. `WHERE` with aggregate functions; pattern matching with `LIKE`.

4. Operating Systems (OS)

~8–10 Questions

- **Process & Concurrency:** Context switching; Dual-mode operation (User vs. Kernel mode); Inter-process communication (IPC); Monitors and Semaphores; Classical problems (Dining Philosophers, Producer-Consumer).
- **Memory Management:** Fixed partition fragmentation; Paging (Page Table mapping virtual to physical addresses); Segmentation (variable-sized blocks); Memory overlays for resource-constrained systems.
- **Storage & Subsystems:** Buffer cache utility; I/O subsystem; File manipulation operations; File systems (FAT - File Allocation Table, NTFS - New Technology File System).

5. Computer Networks & Cybersecurity

~15–18 Questions

- **Network Architecture:** OSI 7 layers; TPDU in Transport Layer; Network Topologies (Bus, Star, Ring, Mesh); Transmission media (Cat-5, Cat-6 UTP cables); Hardware roles (Gateways, Repeaters, Network Interface Cards).
- **Protocols & Addressing:** Data Link framing & contention-based MAC (Ethernet); ARP (mapping IP to MAC); IPv4 datagram size (up to 65,535 bytes); IPv6 length (128 bits); Subnetting; Transport layer (TCP sliding window vs. UDP connectionless); VoIP; DNS; FTP/SFTP; SMTP; HTTP/HTTPS.
- **Cybersecurity:** Packet filtering Firewalls; Public-Key/Asymmetric Encryption (two-key systems: public & private); Symmetric algorithms (AES, DES); CIA Triad (Confidentiality, Integrity, Availability); SQL Injection attacks; Biometrics classification.

6. Theory of Computation & Automata (TOC)

~6–8 Questions

- **Chomsky Hierarchy:** Type 0 (Recursively Enumerable / Turing Machine), Type 1 (Context-Sensitive), Type 2 (Context-Free / PDA), Type 3 (Regular / Finite Automata).
- **Automata Models:** DFA design for divisibility (binary/octal divisible by 3); NFA to DFA conversion (2^n state bound); CFG ambiguity (multiple parse trees); Pushdown Automata; Turing Machine halting states (Accept/Reject).
- **Closure Properties:** Regular language closure under Union, Concatenation, and Kleene Star; Non-closure properties of CFG under intersection and complementation.

7. OOP & Programming (Python, C++, Web)

~12–15 Questions

- **Object-Oriented Programming:** Encapsulation, Abstraction, Polymorphism (Compile-time Overloading vs. Runtime Overriding); Abstract classes vs. Interfaces; `super` keyword; Constructors and Destructors.
- **Python Programming:** Slicing (`L[::-1]`); String operations (`.find()` , `.split()` , `.index()`); Collections (Lists, Dictionaries as mapped types); Exception blocks (`try-except-finally`); Variable scope (`global`); Pandas DataFrames.
- **Web Technologies & Scripting:** HTML semantic tags; CSS presentation rules; JavaScript DOM traversal (`createElement()`); Asynchronous requests via Fetch API/AJAX; XML syntax and DTD entity references.

8. Emerging Technologies (IoT, AI, E-Commerce)

~10–12 Questions

- **Internet of Things (IoT):** Kevin Ashton coining IoT; Analog vs. Digital sensors; Communication standards (ZigBee PAN architecture, BLE - Bluetooth Low Energy, MQTT); SCADA architecture levels; RFID module pinouts (MISO - Master In Slave Out); Industry 4.0 pillars.
- **Artificial Intelligence (AI):** John McCarthy (Father of AI); AI languages (LISP, Prolog); State-space search (Uniform Cost Search, Greedy Best-First, A* search, Min-Max in game trees); Knowledge representation (First-Order Predicate Logic, Semantic Networks, Frames); Expert system components (Knowledge Base, Inference Engine, User Interface).
- **Multimedia & E-Commerce:** Digital audio sampling (8000 samples/sec at 8 bits/sample); Jitter definition in streaming; Business models (B2B, B2C, C2B); Reverse Auctions (buyers request, suppliers compete downwards); Horizontal vs. Vertical e-marketplaces.

Part II: Art of Teaching / Pedagogy (30 Marks)

Essential Scoring Area

Core Module	Key Concepts Tested in Question Papers
Teaching Methods	Heuristic method (H.E. Armstrong - "to investigate"), Socratic/Question-Answer method, Project-based learning, Discussion method, Lecture-demonstration comparisons.
Micro-Teaching & Planning	Micro-teaching cycle (Stanford / Dwight Allen; practiced during teaching training), Lesson plan components, 5E Instructional Model (Engage, Explore, Explain, Elaborate, Evaluate).
Taxonomies & Objectives	Bloom's Domains: Cognitive (Knowledge, Comprehension, Evaluation), Affective (Krathwohl - values/characterization), Psychomotor (physical action/assembly verbs).
Educational Psychology	Howard Gardner's Multiple Intelligences, Piaget's Cognitive Development stages, Attribution theory (Internal vs. External attribution of failure), Progressive Education.
Evaluation & Assessment	Formative (during instruction) vs. Summative (end of unit) evaluation, Diagnostic testing, Continuous and Comprehensive Evaluation (CCE), Multiple-choice test items (Stem & Distractors).

Part III: General Knowledge & Aptitude (20 Marks)

General Scoring Area

- **Logical Reasoning (8–10 Qs):** Letter coding (reverse alphabet pairs, shifted codes), Blood relations ("son of her mother's son"), Direction and distance walks, Number and alphanumeric series completion, Odd-one-out categorization.
- **Quantitative Aptitude (4–6 Qs):** Mensuration (Euler's Polyhedron Formula: $V - E + F = 2$; 3D shapes; triangular numbers), Arithmetic (LCM/HCF word problems, averages, percentages, unit conversions).
- **General Knowledge & Environment (4–6 Qs):** Indian Modern History (Champaran Satyagraha, Swaraj Party, Simon Commission, Vernacular Press Act); Indian Geography (IST longitudes, ports, national highways); General Science (macronutrients, energy types, Kyoto Protocol).